Maths IA-1 — THE ONLY SHEET (one night, one source, final)

> **CLOSE EVERYTHING ELSE.** The Master Worksheet and the Do-or-Die sheet are retired — they were versions of this. THIS file is your entire night and your entire exam prep. It contains: schedule · 27 quick traps · 8 hard questions · 4 application questions · exam playbook · formula card. Nothing outside this file will help you more than what is inside it.

Format-proof · marks-proof · twisted central-team level · target 70%+ · answers inline

FORMAT WARNING: IA-1 layout is UNKNOWN and MARKS are UNKNOWN — the two contests ran differently, and the central team can change format AND weightage any time (MCQ, fill, matching, worded, long-working, any mix, any marks). So this sheet assumes NOTHING about the paper and trains two things only: **quick-mark traps** (become the short questions whatever they're called) and **hard working questions** (become the big questions whatever they're called and however many marks they carry).

THE PACT — the only math that survives an unknown paper: every maths paper, any format, pays two kinds of marks: fast safe ones and slow heavy ones. **70% = near-perfect on the fast ones + 2–3 heavy ones solved fully.** Whether short marks are worth 1 or 3 and long ones 4 or 8, that ratio still lands you at/above 70%.

NIGHT SCHEDULE — follow the clock, no decisions needed

Block	Time	Do
I	22:00 – 00:30	TIER 1 below: 27 quick-mark traps, answers as you go
—	00:30 – 00:50	break: walk, water, food
II	00:50 – 03:40	TIER 2 (8 hard) + TIER 3 (4 applications), full steps
III	03:40 – 04:15	re-solve EVERY wrong one with the answer covered
IV	04:15 – 05:15	memorise FORMULA CARD + PLAYBOOK
V	05:15 – 06:15	timed re-run of Tier 1, 20 min, zero errors
VI	06:00 – 06:45	timed re-run of 5 best Tier 2, 30 min
VII	06:45 – exam time	formula card final glance, get ready, leave early

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TIER 1 — 27 quick-mark traps (these ARE the safe marks in any format)

Each one is designed to be asked as MCQ, fill-in, match-pair or 2-mark short. Learn the trap, not the shape.

1. If $a < b$, which need NOT be true? (a) $a-3 < b-3$ (b) $-4a > -4b$ (c) $a^2 < b^2$ (d) $3a < 3b$

→ **(c)** — $a = -5$, $b = 1$ → $25 < 1$ false. Everything that "flips" or "squares" is the bait.

2. $\log_2(x - 1) = 3 \rightarrow x = ?$

→ **9** ($x - 1 = 8$; the trap answer is 8)

3. $x^2 + (k - 3)x + k = 0$ has equal roots → sum of possible k = ? product = ?

→ **sum 10, product 9** — $D = k^2 - 10k + 9 = 0 \rightarrow k = 1$ or 9. Underline SUM vs PRODUCT before answering.

4. Solution of $(x - 3)/(x + 1) < 0 \rightarrow$

→ **$-1 < x < 3$** (opposite signs between the roots; $x = -1$ excluded)

5. Digits of 2^{100} ($\log 2 = 0.3010$) →

→ **31** ($\lfloor 30.10 \rfloor + 1$)

6. Last two digits of $7^{200} \rightarrow$

→ **01** ($7^4 = 2401 \equiv 01 \bmod 100$)

7. gcd(a,b) = 12, lcm(a,b) = 120 → ordered pairs (a,b) = ?

→ **4** — $a = 12x$, $b = 12y$, $xy = 10$, $\gcd(x,y) = 1$: (1,10)(2,5)(5,2)(10,1)

8. "All multiples of 4 are even. 12 is even. So 12 is a multiple of 4." Flaw?

→ **even ≠ multiple of 4 (6 is even, not a multiple of 4)** — reasoning ran backwards

9. $|x| = -x$ exactly when →

→ **$x \leq 0$** (0 included — trap answer: $x < 0$)

10. $x + 1/x = 4 \rightarrow x^3 + 1/x^3 = ?$

→ **52** ($4^3 - 3 \cdot 4$)

11. $\log_5 125 + \log_2 32 - \log_{10} 10000 = ?$

→ **4** ($3 + 5 - 4$)

12. Remainder of $10^{10} \div 7 = ?$

→ **4** ($10 \equiv 3$; $3^6 \equiv 1$; $10 = 6+4 \rightarrow 3^4 = 81 \equiv 4$)

13. Divisors of 360 = ?

→ **24** ($2^3 \cdot 3^2 \cdot 5 \rightarrow 4 \cdot 3 \cdot 2$)

14. $(x - 2)$ divides $x^3 - kx^2 + 10x - 8 \rightarrow k = ?$

→ **5** ($f(2) = 20 - 4k = 0$)

15. 4^(x+1) = 8^x → x = ?

→ **2** ($2x + 2 = 3x$)

16. Product of roots of $3x^2 - 7x + 2 = 0 \rightarrow$

→ **2/3** (c/a)

17. **MATCH SET:** a) $x + 1/x = 3 \rightarrow x^2 + 1/x^2$ · b) lcm(12,18) · c) 100 mod 7 · d) $\log_9 27$ · e) divisors of 60 — against: 7 / 36 / 2 / 3/2 / 12

→ **a-7 · b-36 · c-2 · d-3/2 · e-12**

18. **Complete the square:** $x^2 - 8x + 5 \rightarrow$ **$(x - 4)^2 - 11$** (halve −8, square 16, subtract back)

19. **Euclidean algorithm:** gcd(1071, 462) with steps → $1071 = 2 \times 462 + 147$; $462 = 3 \times 147 + 21$; $147 = 7 \times 21 + 0 \rightarrow$ **21** (steps ARE the marks)

20. **Counting:** integers 1–100 divisible by 6 OR by 8 → $16 + 12 - 4 =$ **24** (inclusion–exclusion: $\lfloor 100/6 \rfloor$, $\lfloor 100/8 \rfloor$, $\lfloor 100/24 \rfloor$)

21. **For all x:** find k so $x^2 + 2x + k > 0$ for EVERY real $x \rightarrow D = 4 - 4k < 0 \rightarrow$ **$k > 1$** (note $k = 1$ fails: $x = -1$ gives 0)

22. **Factorise fully:** $x^4 - 16$ → **$(x - 2)(x + 2)(x^2 + 4)$** (difference of squares TWICE)

23. **Exponent rules:** $2^3 \cdot 2^5 \div 2^4 =$ → **$2^4 = 16$** (add exponents, subtract on division)

24. **Log laws:** express log 45 using log 3 and log 5 → **2 log 3 + log 5** ($45 = 9 \times 5$)

25. **Log definition:** if $2^x = 7$ then x = → **$\log_2 7$** (= log 7 / log 2, change of base)

26. **gcd × lcm:** gcd(a,b) = 8, lcm(a,b) = 96, a = 24 → b = → **32** ($b = 96 \times 8 \div 24$)

27. **Worded LCM:** buses leave every 15 min and 20 min, both at 8:00 → next together at → **9:00** (lcm = 60)

TIER 2 — 8 hard questions (the 10-mark engine; exam needs 2–3 of these perfect)

Full-mark rule everywhere: given → formula → steps → answer → one sentence of meaning. Any format that asks for "working/reasoning" pays per line.

H1. Last two digits of 31^{31}.

→ $(30 + 1)^{31} = 1 + 31 \times 30 +$ terms with $900 \equiv 0 \pmod{100}$ → $1 + 930 = 931$ → **31**

H2. Prove $n^4 - n^2$ divisible by 12 for every integer n.

→ $n^2(n-1)(n+1)$: three consecutive integers give factor 3; n even → n^2 gives 4, n odd → both neighbours even give 4. Divisible by 3 and 4, coprime → **divisible by 12** ∎

H3. Greatest number dividing 552, 967, 1032 leaving the SAME remainder?

→ differences: 415, 65, 480 → Euclid on (415, 65): $415 = 6 \times 65 + 25$, $65 = 2 \times 25 + 15$, $25 = 15 + 10$, $15 = 10 + 5$ → **5**; check remainders 2, 2, 2 ✓

H4. Boat: same distance upstream in 4 h as downstream in 2 h, stream 3 km/h. Boat speed & distance?

→ $4(b-3) = 2(b+3)$ → **b = 9 km/h**, distance **24 km** (check: $2(9+3) = 24$ ✓)

H5. Solve $\log x + \log(x + 3) = 1$.

→ $x^2 + 3x - 10 = 0$ → x = 2 or –5 → **x = 2** (must write: –5 rejected, domain $x > 0$)

H6. Solve $5x \equiv 3 \pmod{8}$.

→ inverse of 5 is 5 ($25 \equiv 1$) → $x \equiv 15$ → **$x \equiv 7 \pmod{8}$**

H7. Four bells toll together at noon, then every 6, 8, 12, 15 s. How many times together in 1 hour including start?

→ lcm = 120 s; 3600/120 = 30 → **31 times**

H8. Sum of a number and its reciprocal is 10/3. Find it.

→ $3x^2 - 10x + 3 = 0$ → $(3x-1)(x-3)$ → **x = 3 or 1/3** (both — word problems can have two)

TIER 3 — 4 application questions (central teams LOVE worded problems)

Same rule: given → formula → steps → answer → one sentence of meaning.

T1. (algebra → quadratic) A garden is 5 m longer than wide; area 36 m². Dimensions? Then same garden widened to 40 m² — new width?

→ $w(w + 5) = 36 \rightarrow w^2 + 5w - 36 = 0 \rightarrow (w + 9)(w - 4) = 0 \rightarrow$ **w = 4 m, l = 9 m** (reject –9)

→ $w^2 + 5w - 40 = 0 \rightarrow w = (-5 + \sqrt{185})/2 \approx$ **4.3 m** ($\sqrt{185} \approx 13.60$)

T2. (inequality chain) IQ = MA/CA × 100. Children aged 15 with 90 ≤ IQ ≤ 150 — mental age range?

→ MA = IQ × 15/100 → 90→13.5, 150→22.5 → **13.5 ≤ MA ≤ 22.5 years**

T3. (rocket — the full 4-mark shape) $h(t) = -5t^2 + 20t + 25$: (i) zeros (ii) max height without calculus (iii) h(6) = –35 — "35 m underground"?

→ (i) $-5(t - 5)(t + 1) \rightarrow$ **t = 5 s** (t = –1 rejected: time ≥ 0)

→ (ii) vertex $t = -b/2a = 2 \rightarrow h(2) = -20 + 40 + 25 =$ **45 m**

→ (iii) **NO** — formula is only valid during flight; t = 6 s is past landing, negative height is extrapolation outside the model

T4. (log application) $pH = -\log_{10}[H^+]$. A: $[H^+] = 10^{-3}$, B: $10^{-4.5}$. Which more acidic, by how much?

→ pH A = 3, pH B = 4.5 → lower pH = more acidic → **A more acidic by 1.5 pH units**

PLAYBOOK — extracting 70% from an unpredictable, confusing, unknown-marks paper

Universal order (90 min):

1. **First 5 min:** read the WHOLE paper, note what formats exist today (they differ!), count marks per part. No solving yet.

2. **Next pass (first ~⅓ of your time):** every question answerable in ≤ 90 seconds — values, fills, one-liners, obvious MCQs. This is your safe bank. Do it in ANY order.

3. **Matching (if present):** pair independently, lock 3+ pairs minimum.

4. **Remaining (~half your time):** the heavy ones. First pick 2–3 whose chapter you recognise, solve fully with steps. Then fight the rest.

5. **Last ~10%:** every blank gets SOMETHING; check signs, rejections, units.

70% scoreboard mid-exam: after the safe pass, ask: 'did I lock the easy ones and fully solve two heavy ones?' If yes → you're on track, no panic. If no → the heavy ones are where you go next, never sideways into new territory.

When the question confuses you (twist protocol):

- Read twice, **underline exactly what is asked** (sum? product? value of k? "need NOT"? "always"?).
- Name the chapter, write the givens, write the one formula that fits.
- Lost anyway? Write formula + substitution + anything derivable → partial credit is real credit wherever marks exist.

Format-switch rules (whatever they throw):

- MCQ with no stated negative marking → **never leave blank**; eliminate 2 first.
- "Explain/justify/why" → full sentence, name the reason (domain, flip, sign, rejection, converse).
- Interpretation ("is this correct?") → YES/NO + reason, even if unfinished.
- Two candidate answers → plug both into the ORIGINAL statement; original always wins.
- Unknown/freak format → it's still chapters 1–6 underneath: identify which, apply formula card.

3-minute rule: not cracked in 3 min → abandon, flag, return at minute 75.

FORMULA CARD — memorise in block IV, final glance at VII

Ch1: $(x+1/x)^2 - 2$ · $(x+1/x)^3 - 3(x+1/x)$ · $(a+b)^2 - 2ab$ · $4ab$ identity · digits = $\lfloor \log \rfloor + 1$ · area = big shape − removed pieces

Ch2: $D = b^2 - 4ac$ (touch $\iff 0$, none $\iff <0$) · sum $-b/a$, product c/a · vertex $-b/2a$ · both roots + $\iff$ sum>0, product>0, $D \geq 0$ · equal roots $\iff D = 0$

Ch3: flip only on ×÷ negative · $|x| \leq a \iff -a \leq x \leq a$ · quadratic < 0 between roots · rational: sign chart, exclude denominator · "all x" $\iff D<0$ & $a>0$

Ch4: + logs multiply, − divide · $\log_a a = 1$ · $a^b = N \iff b = \log_a N$ · chain collapse · digits formula · **domain first, reject negatives/zero**

Ch5: gcd×lcm = product · Euclid steps to 0 · same remainder $\Longrightarrow$ gcd of differences · $\tau = \Pi(e_i+1)$ · or-count = A + B − both · pair form gcd·x, gcd·y with $\gcd(x,y)=1$

Ch6: reduce base mod n → reduce exponent mod cycle · last-digit cycles len 4 · mod 100 heroes: $7^4 \equiv 01$, $3^{20} \equiv 01$ · inverse = multiply to $\equiv 1$ · $ax \equiv b$: divide by gcd first · consecutive-product proofs

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This is the whole plan. Tiers 1-2-3 in order, then playbook, then formula card. Nothing else exists tonight.